



HUMAN MOVEMENT 1 (BASIC ANATOMY IN SPORTS AND EXERCISE)

ACADEMIC

Grade 11/12

Curriculum Guide

HUMAN MOVEMENT 1
(BASIC ANATOMY IN SPORTS AND EXERCISE)
Prerequisite: None

Course Description:

This course covers the fundamental structures and functions of the human body. Learners will explore key anatomical systems, including the skeletal, muscular, cardiovascular, nervous, and respiratory systems, among others. Emphasis is placed on understanding the relationships among different body parts and their roles in maintaining overall health and functionality, thereby developing a foundational knowledge essential for careers in health sciences.

Elective: Academic

Time Allotment: 80 hours for one term

Content Standard:

- The learners demonstrate understanding of the basic structure and function of the human body in relation to fitness, exercise, and health for the optimization of human movement performance.

Performance Standard:

- The learners evaluate basic functional movement patterns using anatomical knowledge to optimize performance across different needs and skill levels.

LEARNING COMPETENCIES	CONTENT						
The learners... 1. explain the basic structure of the human body;	Human Structure and Processes • Foundations of Human (Gross Structure) • Foundations of Human Physiology (Function)						
2. describe the function and interaction of body systems in relation to fitness and exercise;	<table border="1"> <thead> <tr> <th data-bbox="1043 399 1424 504">Major Components and Functions of the Human Movement Systems:</th><th data-bbox="1424 399 2087 504">Body Responses and Adaptations to Exercise</th></tr> </thead> <tbody> <tr> <td data-bbox="1043 504 1424 544"> • The Nervous System </td><td data-bbox="1424 504 2087 718" rowspan="3"> • Balance, Stability, and Coordination • Heart Rate and Exercise Response • Fatigue • Energy Systems • Growth and Maturation </td></tr> <tr> <td data-bbox="1043 544 1424 649"> • The Cardiorespiratory System </td></tr> <tr> <td data-bbox="1043 649 1424 718"> • The Musculoskeletal System </td></tr> </tbody> </table>	Major Components and Functions of the Human Movement Systems:	Body Responses and Adaptations to Exercise	• The Nervous System	• Balance, Stability, and Coordination • Heart Rate and Exercise Response • Fatigue • Energy Systems • Growth and Maturation	• The Cardiorespiratory System	• The Musculoskeletal System
Major Components and Functions of the Human Movement Systems:	Body Responses and Adaptations to Exercise						
• The Nervous System	• Balance, Stability, and Coordination • Heart Rate and Exercise Response • Fatigue • Energy Systems • Growth and Maturation						
• The Cardiorespiratory System							
• The Musculoskeletal System							
3. execute basic movement patterns using different anatomical terms and movement descriptors;	Anatomical Terms • Definition and Importance of Anatomical Terms • Standard Anatomical Position • Planes of Motion (Sagittal, Frontal, Transverse) • Axes of Rotation (Mediolateral, Anteroposterior, and Longitudinal) Directional Terms in Anatomy • Superior vs. Inferior • Anterior (Ventral) vs. Posterior (Dorsal) • Medial vs. Lateral • Proximal vs. Distal • Superficial vs. Deep Movement Descriptors and Joint Actions • Linear and Angular Movements • Primary Joint Movements ○ Flexion vs. Extension ○ Abduction vs. Adduction ○ Rotation (Internal and External)						

LEARNING COMPETENCIES	CONTENT
4. analyze the various movement patterns of the upper extremity segments;	Movement Analysis of the Upper Extremities • Shoulder • Elbow • Wrist • Hand • Finger
5. analyze the various movement patterns of the lower extremity segments; and	Movement Analysis of the Lower Extremities • Hip • Knee • Ankle • Foot
6. execute basic movement patterns using different anatomical terms and movement descriptors.	Analysis of Basic Movement Patterns • Walking • Running • Jumping • Throwing

GLOSSARY

<i>Anatomical Terms</i>	The language of anatomy used to describe the location, orientation, and movement of body parts and structures.
<i>Autonomous Stage</i>	The final stage in which movements are accurate, consistent, and efficient. The learner can perform the movement with little or no cognitive processing
<i>Balance</i>	The ability to maintain the body's center of gravity within its base of support (BOS). It may also be described as the ability to maintain equilibrium, a condition in which all acting forces are balanced, resulting in a stable system.
<i>Cardiovascular System</i>	The body system responsible for transporting oxygen, nutrients and hormones throughout the body and eliminating cellular metabolic waste
<i>Coordination</i>	The ability to activate the right muscle at the right time and with appropriate intensity to achieve effective movement.
<i>Directional Terms</i>	Terms used to describe the positions of structures relative to other structures or locations in the body.
<i>Energy Systems</i>	The physiological systems that produce and supply energy for physical activity.
<i>Fatigue</i>	Physical and/or mental weariness resulting from exertion, characterized by an inability to continue exercise at the same intensity and a resulting decline in performance.
<i>Functional Anatomy</i>	It is a field that uses basic structural knowledge of human anatomy to develop an understanding of the functional significance of the structures of the musculoskeletal system.
<i>Growth</i>	A change in body size, body composition, or dimensions of a specific region of the body (Joyce and Lewindon, 2014).
<i>Heart rate</i>	It refers to the number of times the heart beats per minute and is directly related to the workload being placed on the heart.

<i>Human Anatomy</i>	It is often defined as the study of structures in the human body. Anatomy focuses on the description of form, or how body structures at different levels look.
<i>Human Movement</i>	It is the movement produced by the human body due to the contraction of muscles and bending of bone joints. Human movements are controlled by the nervous system. Hence, the human movement incorporates the use of muscles, ligaments, joints, and bones. Kinesiology is the field that deals with the study of human movements.
<i>Human Movement Analysis</i>	It is the study of human motion through the use of various techniques such as biomechanics, kinesiology, and motion capture technology. A method of analysis refers to the systematic approach used to study, interpret, and evaluate a subject, phenomenon, or process.
<i>Human Movement System</i>	It is a system of physiological organ systems that interact. to produce movement of the body and its parts.
<i>Human Physiology</i>	It studies the “nature” of the human body, nature in the sense of how structures at different levels work. Physiology focuses on function, or how structures at different levels work.
<i>Lower Extremities</i>	It refers to the part of the body that includes the hip, thigh, knee, leg, ankle, and foot. It consists of bones, muscles, tendons, ligaments, nerves, and blood vessels that work together to provide support, stability, and mobility for daily activities.
<i>Maturation</i>	It is the progression of the human body towards adulthood. The rate and timing of maturation can be highly variable, and even different systems can mature at different times. For example the skeletal system will mature at a much later date than the reproductive system (Lloyd and Oliver, 2014).
<i>Movement Assessment</i>	It evaluates a person's ability to perform functional movements, often focusing on mobility, stability, strength, coordination, and overall movement patterns. It is used in various fields, such as physical therapy, sports science, and fitness training, to identify movement deficiencies, imbalances, or risks of injury.
<i>Musculoskeletal System</i>	It is a Mechanical support, posture and locomotion.

<i>Skeletal system</i>	The body system composed of bones and cartilages.
<i>Stability</i>	The ability of the body to maintain postural equilibrium and support joints during movement.
<i>Torso</i>	The main part of the body contains the chest, abdomen, pelvis, and back. Most of the body's organs and the backbone are found in the torso. It is also called the trunk.
<i>Upper Extremities</i>	The upper limbs extending from the shoulder girdle to the fingers, including the muscles, ligaments, and joints of the shoulder, elbow, wrist, and hand.
<i>Voluntary movement</i>	Intentional and goal-directed actions initiated and controlled by the higher centers of the brain, particularly the motor cortex, basal ganglia, and cerebellum.

REFERENCES

BPT, U. T. (2024, June 24). *Human anatomy terminology*. Muscle and Motion.

<https://www.muscleandmotion.com/human-anatomy-terminology/>

Contemporary issues and theories of motor control, motor learning, and neuroplasticity. (2015, April 9). *Clinical Gate*.

<https://clinicalgate.com/contemporary-issues-and-theories-of-motor-control-motor-learning-and-neuroplasticity/>

Emmert, W. (2010). *Functional movement screen*. American College of Sports Medicine.

<https://www.acsm.org/docs/default-source/regional-chapter-individual-folders/northland/nacsm--wes-e--fms9a9b0c1f5032400f990d8b57689b0158.pdf>

Energy system. (n.d.). *Open Risk Manual*.

https://www.openriskmanual.org/wiki/Energy_System

Evans, W. J., & Lambert, C. P. (2007). Physiological basis of fatigue. *American Journal of Physical Medicine & Rehabilitation*, 86(1), S29–S46.

<https://doi.org/10.1097/phm.0b013e31802ba53c>

Gross and fine motor skills. (n.d.). *Early Movers*.

<https://www.earlymovers.org.uk/motor-skills>

Growth, maturation and development. (n.d.). *World Rugby Passport*.

<https://passport.world.rugby/conditioning-for-rugby/introduction-to-conditioning-children/long-term-athlete-development/growth-maturation-and-development/>

Human movement. (2021, October 22). *Bartleby*.

<https://www.bartleby.com/subject/science/biology/concepts/human-movement>

Latash, M. L., Levin, M. F., Scholz, J. P., & Schöner, G. (2010). Motor control theories and their applications. *Motor Control*, 14(2), 1–43.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3017756/>

Mobility and stability: Joint functions when we move. (n.d.). *National Academy of Sports Medicine*.

<https://blog.nasm.org/certified-personal-trainer/mobility-and-stability-joint-functions-when-we-move>

Motor control and motor learning. (2018, April 26). *Trek Education*.

<https://exercise.trekeeducation.org/motor-learning/>

Sahrmann, S. (n.d.). *The human movement system: Our professional identity*. Program in Physical Therapy, University of Toledo.

https://www.utc.edu/sites/default/files/2020-12/annualforum_2017_by_sahrmann_012617.pdf